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Wind farm acceptance for sale? Evidence from the Danish wind farm co-ownership scheme

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ABSTRACT

The Danish Renewable Energy Act features several financial incentive structures with direct local level implications. One of these is the wind farm co-ownership scheme, WCS. In this study, we explore local perceptions of the WCS via survey data collected during the Danish near-shore bid for tender in 2015 from almost 2000 respondents. Empirical evidence suggests that demographic facts, such as gender and age, influence the general appeal of the scheme, and as engagement in wind farms via the WCS presupposes investment liquidity, the WCS is not equal for all. Furthermore, most potential WCS-investors already support the planned wind farm projects, and many project opponents will not engage themselves in something they are against in principle. Finally, economic benefits potentially gained via the WCS do not appear to compensate for values feared violated by wind farms by many wind farm project stakeholders. While the WCS is a very positive policy attempt at creating local engagement via wind farm co-ownership, it is clear that the scheme alone will not adequately compensate for local wind farm related grievances. Real world facts and complications, such as demographics, preconceived project perceptions and personal values, get in the way. Implications of the study for related policies are discussed.

1. Introduction

1.1. The Danish history of wind

Denmark is a wind nation. Located in a region of the world with rich, natural wind resources, historically the small nation has successfully utilized this readily available natural source of energy for multiple purposes (Nissen et al., 2009). Currently about 40% of national electricity consumption is based on wind power (Bæk et al., 2016; Dansk Energi, 2015; Quartz + co, 2015), and wind energy remains a corner-stone in national ambitions for the transition to sustainable energy resources (Danish Ministry of Climate Energy and Building, 2012; Sovacool, 2013).

The wind industry in Denmark has deep-seated historical roots: From grain-grinding wind-mills to Poul la Cour, a physicist with who had a dream that the entire rural population in Denmark would receive their energy from small, local power-plants fueled by wind, through anti-nuclear idealists seeking alternatives, to the modern, industrial-scale wind-turbine parks of today characterized by individual and/or corporate ownership (Danmarks Vindmølleforening, 2013; Nissen et al., 2009). Development of wind energy, and the broader wind energy related supply chain, has been politically prioritized in Denmark (Quartz + co, 2015, pp. 35–36; Sovacool, 2013). Despite this national history of wind, however, as the size and costs of turbines grew, and as larger companies gradually replaced individually owned turbines and smaller cooperatives as key players, the wind farm opposition also grew (Warren and McFadyen, 2010). Now it seemed that not all local populations, and potential wind farm neighbors, shared quite the same enthusiasm for wind power related technological advancements, and plans for more, as planners and policy-makers did. This emergent local opposition (Sovacool, 2013; Warren and McFadyen, 2010) was not well in sync with Danish government ambitions for a grand scale transition to a largely wind energy driven sustainable energy infrastructure. Something had to be done.

The Danish Renewable energy Act of 2009, REA, can be seen as a policy answer to the energy transition challenges posed by local wind farm resistance. REA features four policy measures with the aim of promoting “local support for the establishment new wind farms” (Olsen and Anker, 2014, p. 146), and it does so in the form of financial incentive structures for windpower with direct local level implications (Sperling et al., 2010, p. 5445). These are the wind farm co-ownership

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scheme, the community benefit scheme (the green scheme),[1] the loss of property-value scheme and the guarantee fund (Anker and Jørgensen, 2015; Olsen and Anker, 2014).[2,3] A popular view has been that greater provision of benefits to communities affected by wind farms would make wind farms "more socially acceptable" (Cowell et al., 2011, p. 552). And yet, as we argue in this paper, the potential mitigating and compensating effects of community benefit schemes may be quite a complicated matter, mirroring inherent community heterogeneity, values, contextual complexity, issues of process, power, justice and trust (Bidwell, 2013; Cass et al., 2010; Cowell et al., 2011; Goedkoop and Devine-Wright, 2016; Kerr et al., 2017; Warren and McFadyen, 2010). Thus, the pending question remains what the local effects of the chosen community benefits schemes, CBSs, actually are.

1.2. The Danish wind farm co-ownership scheme

In this study, we examine the potential local effects of one of the four Danish REA policy measures, namely the option to purchase wind farm shares scheme, or the wind farm co-ownership scheme (WCS). WCS is a community benefit scheme (Cowell et al., 2011; Kerr et al., 2017); it is an investment and co-ownership scheme for citizens living within pre-defined geographical proximity of windfarms. Minimum 20% of the wind farm must be offered to the public as wind farm shares at cost-price.[4] WCS then redistributes potential wind farm investment returns among WCS investors, and it does so following a standardized investment procedure defined through the legal framework of REA (Anker and Jørgensen, 2015; Energi-, Forsynings- og Klimaministeriet, 2017; Energinet.dk, 2012). Accordingly, the WCS can be seen as an "interventionist" statutory policy approach (Kerr et al., 2017) to mitigating public wind farm opposition/promoting local project acceptance. The scheme has been internationally recognized as a serious policy attempt at creating local wind farm project engagement through co-ownership (Aitken et al., 2014; Ellis and Ferraro, 2017; Kerr et al., 2017; National Economic and Social Council & SLR Consulting, 2015).

The aim of this study is to understand how, and to what extent, the WCS as a CBS potentially promotes local engagement in and acceptance of planned near-shore wind farms among potential near-shore wind farm 'neighbors' in Denmark.[5] More specifically, we pursue evidence as to how the WCS is perceived among the target groups of the scheme. Survey data for the study, collected during the Danish near-shore bid for tender in 2015, mirrors perceptions of the WCS as indicated by almost 2000 respondents.

We lean on two main theoretical anchors: 1) Reflecting the redistributive ambition and logic of the WCS we turn to theoretical insights from the field of environmental justice, specifically distributive justice (Goedkoop and Devine-Wright, 2016; National Economic and Social Council & SLR Consulting, 2015; Walker, 2012). 2) Taking the analysis to the individual, value based level, we are inspired by the theory of "webs of significance" (Geertz, 1973). This theory aptly elucidates the potential mental reach, as it were, of the WCS within our webs of personalized significance; it gives us an indication of the extent to which the WCS may actually comfort in places where personal values are perceived violated via wind farm planning and development.

Our investigation is guided by three main research questions:

1. What demographic factors matter for willingness to invest in the WCS?
2. To what extent does the WCS appeal to wind farm opponents?
3. To what extent is some level of distributive justice achieved through the WCS?

The paper is organized as follows: In Section 2, we briefly introduce theoretical concepts and background. In Section 3, the data collection process, relevant sample characteristics and survey questions are described, and the key data tendencies are presented. In Section 4, a logistic regression controls for the effects of demographics, general support for wind energy and indicated attitude towards the specific plans for local near-shore wind farms on willingness to invest in the WCS. Finally, open-ended respondent comments provide qualitative depth to our findings. Informed by these powerful empirical insights, in Section 5 the paper concludes with a discussion of study implications for related policy.

2. Background and theory

2.1. Procedural justice, distributive Justice and place related impact

Recent research on public perceptions of wind farms has unveiled in-depth "nuances and realities of public [wind farm] opposition and support. (….) This growing body of literature points to the complexities of public opinions (…), to the importance of considering local values and contextual factors (…), [and generally] to the multiple forms that responses to wind farms can take (Aitken et al., 2016, p. 558)". People/place relations have been identified as important for local perceptions of wind farms/renewable energy technologies, RETs (Devine-Wright, 2009; Ellis and Ferraro, 2017; Johansen,), and person's/peoples' particular attachments to certain places may shape their reactions towards RET-related change to those places. Some key issues identified as important for local perceptions of/attitudes towards wind farms are concerns about project impact on the local (coastal) landscape, the environment, biodiversity, noise, health, tourism, and concerns about project impact on the future local well-being generally (Devine-Wright, 2009; Devine-Wright and Howes, 2010; Ellis and Ferraro, 2017; Pasqualetti, 2011; Zaunbrecher and Ziefle, 2015).

Based on European case studies two main types of locally perceived injustice associated with wind farm planning and development have been identified. 1) Procedural (in)-justice concerns the degree to which local wind farm related decision-making processes are perceived as open and fair by the local public, and relating to this, that the projects

[1] The community benefit scheme/green scheme supported projects strengthening landscape and recreational values, information- and cultural activities that promoted the acceptance and use of renewable energy resources in the wind farm host municipality. The green scheme only applied for onshore wind farm projects, and it expired on the 20th of February 2018 (Anker and Jørgensen, 2015; Energinet.dk, 2018). While the green scheme was designed as a true community benefit scheme, the funds from this scheme are largely left untouched. Reasons for this are yet unknown, but it may have to do with lacking knowledge of the green scheme locally and/or a somewhat complicated application process.

[2] The guarantee fund supports preliminary project investigations for local wind farm initiative groups through a loan guarantee (Anker and Jørgensen, 2015).

[3] Ideally all of the relevant wind power related policy incentive schemes should be evaluated in-depth and in combination, but the scope of such potential research would far exceed what can be presented within the format of a single journal article.

[4] Citizens eligible for investing in wind farm shares through the WCS must be 1) minimum 18 years of age. Citizens must also be registered in the Danish CPR-register with 2) permanent residency address a) up to 4.5 km from the wind farm project site, or b) in the municipality hosting the wind farm project, or *c*) in municipalities with coastline up to 16 km from off-shore project sites when shares are sold. Shares are sold at cost price. Group a is prioritized, and citizens in this group a can buy up to a maximum of 50 wind farm shares. The remaining shares are then offered to citizens above 18 in groups b and c. The wind farm developer disposes freely over shares not sold during the min 8 week period when shares are offered for sale (Anker and Jørgensen, 2015; Energinet.dk, n.d.b).

[5] In this context the Danish Energy Agency tender criteria define 'near-shore' as minimum 4 km from the coast (Danish Energy Agency, 2013).

are often "foisted on a local community rather than accepted by choice" (National Economic and Social Council, & SLR Consulting, 2015, pp. iv–viii). 2) Issues of distributive justice have also proved crucial. In this paper, we refer to distributive justice broadly as "justice […] conceived in terms of the distribution or sharing out of goods (resources) and bads (harm and risk)" (Walker, 2012, p. 10) from impacts on the local environment. Distributive justice "concerns the ways the distribution of costs, risks, and benefits between different actors is perceived" (Goedkoop and Devine-Wright, 2016). In the context of wind farm planning and development "distributional justice refers to issues emerging from the perception that while local communities (…) bear the main impacts of a wind energy proposal (….) external bodies, particularly developers, accrue most of the benefits" from the wind farm projects (National Economic and Social Council & SLR Consulting, 2015, pp. iv–viii). So some local communities hosting wind farms have experienced a prevailing sense of distributive injustice: the perception that "goods" and "bads" (Walker, 2012, p. 10) from the wind farm projects have, in fact, not been evenly distributed. Rather the opposite. Reflecting the redistributive ambition and logic of the WCS the notion of distributive justice is at the core of our WCS analysis.

2.2. Complexities of community benefits schemes

Social acceptance of renewables has been widely researched, and much attention has been paid to the provision of community benefits in this context. Whatever language is used, "community benefits" in the context of energy developments will be a form of compensation for local impact; a "payment or exchange where there is reciprocity" (Kerr et al., 2017). Research into different models for community benefits in the context of wind farm development concludes that such initiatives are, to say the least, problematic in practice: Some members of the public interpret community benefit initiatives as "bribery" (Cass et al., 2010.), and shared ownership can be complicated due to differing motives/ideologies, risk-willingness and a lack of trust among parties involved (Goedkoop and Devine-Wright, 2016).

The notion of community is intrinsically complex (Cowell et al., 2011, p. 544; Kerr et al., 2017). Consequently, in community engagement activities, critical attention must be paid to who participates, conflicting interests and existing local relationships of power (Aitken et al., 2016). And all-though economic benefit may appeal to some members of the community, it will not necessarily address all value-driven community concerns (Bidwell, 2013, p. 198). Some research has confirmed increasing levels of renewables acceptance via community ownership/co-ownership (Kerr et al., 2017; Musall and Kuik, 2011; Warren and McFadyen, 2010).

As implied, the WCS represents a very real attempt at promoting local engagement in and more widespread social acceptance of wind farm projects. It does so as a CBS, a financial incentive scheme mandated by law (Anker and Jørgensen, 2015; Energi-, Forsynings- og Klimaministeriet, 2017; Kerr et al., 2017; Olsen and Anker, 2014; Sperling et al., 2010) via the redistribution of potential wind farm economic "goods" locally to WCS investors. Thus, potentially WCS will achieve some degree of distributive justice (Walker, 2012) in the context of wind farm planning and development. What we ask, however, is to what extent this scheme actually reaches the core of project-induced grievances amongst members of the public affected by those projects. Do potential benefits from the WCS relieve some of the values activated, and potentially violated (see Bidwell, 2013), by local wind farm planning and development? As we argue, the fact that the WCS is, by nature, an investment scheme rooted in an economic rationality of "goods" via financial benefits ultimately influences the potential mediating effect and "reach" of the scheme. After all: If other personal values than those directly rooted in an economic rationality powerfully influence project perceptions, then the potentially mediating, compensating and economic redistributive effects of the WCS does not necessarily 'soothe where it hurts'.

2.3. Webs of significance: shaping our actions, reactions and perceptions

In order to illustrate potential value-related mitigating effects of the WCS we turn to our second theoretical anchor, the symbolic imagery of Glifford Geertz. Geertz describes culture as "webs of significance" (Geertz, 1973). These constitute "a cohesive narrative of existence, a kind of mental map (…) [serving as] "a guide to the terrain of life" (Nelson, 1996). According to Geertz, the world contains within it multitudes of meanings. Individual subjects, then, must interpret this complexity for a meaningful existence (Hylland Eriksen and Nielsen, 2002). We might go on to say that the different threads in those overarching personalized webs of significance, the webs of interpreted value via which we navigate in life, then come into play and guide/shape our actions, reactions and perceptions related to both concrete, tangible and potential change. The metaphor 'threads of personalized significance' requires a little unpacking: What we imply is that within our internal webs of significance some of our precious values, through which we add meaning to life and existence, run on parallel value tracks. Like the threads in a spiders' web, differing values play their crucial parts in the complex totality of the intricately interwoven webs of value systems that constitute our mental navigational map. But just as in the threads of a spiders web some of these individual value-sets, or personalized threads of significance, will never meet because they run in parallel. Some run near the center. Others run along the periphery of that same circular structure. For example, while one person may appreciate a profitable investment, and economic benefit generally (one value-set), values associated with, and potentially activated by, attachment to a local area may be at a very different place in that person's complex value system (a parallel value-set). These value-sets might be so far apart that they just do not connect. Not all values-sets, the threads of personalized significance, are activated at all times and at all places. Some values are activated by certain events in life. Others will remain passive in those same situations. Some value-sets may remain silent when confronted with a specific theme in very general terms, but may be activated when confronted with that same theme in very specific terms; when personal lives, and locally grounded value-sets, are affected (see Bidwell, 2013). One example could be that an individual supports wind energy on a general level, but actively opposes concrete plans for sustainable energy infrastructure, such as wind farms, locally (Bell et al.,2005, 2013). (Fig. 1)

3. Method

3.1. Survey administration and sample

Survey data was collected during the Danish Energy Agency, DEA, 2015-2016 near-shore wind farm tender process. Data was collected at the time between preliminary project bids for tender and final tender

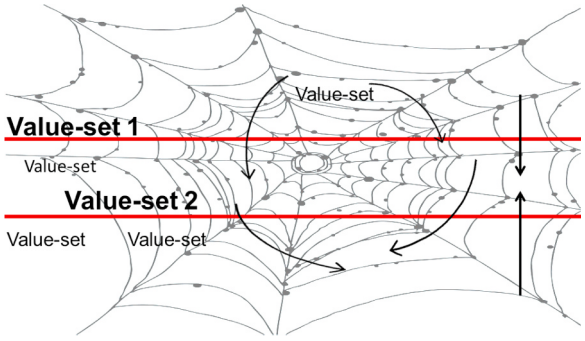


Fig. 1. Webs of significance. Parallel value-sets and personalized webs of significance.

specifications. The randomized sample was evenly distributed by the 5 DEA selected near-shore wind farm sites located off the coast off mainland Denmark. The sample was based on property ownership/residency, and it included both permanent area residents (PRs) and second home owners (SHOs).

Citizens selected for the sample received a written survey invitation and introduction letter posted to their permanent residency address. The introduction letter included printed visualizations of the potential local wind farm projects at the specific sites, and as in the publicly available EIA-material, the printed visualizations demonstrated both 3 MW and 10 MW local near-shore wind farm scenarios. Response-rates from the 5 potential wind farm sites vary little compared to the full sample distribution and the final survey response-rate was close to 2000. Survey research design and content was officially approved by the Danish Data Protection Agency, Datatilsynet.

3.2. Questionnaire

The survey was completed as an e-survey, and most questions were presented in grid format with appropriate Likert-scaling. Key survey questions concerned general attitude towards, and anticipated impact of, the potential near-shore wind farms locally. The survey also touched upon themes related to place and place-attachment, local engagement/sense of responsibility, municipal planning practice, the transition to sustainable energy generally - and topics related to wind farm development specifically. The section on the WCS presented in this paper was also inspired by pre-survey explorative interviews completed after sales of wind farm shares from a small, on-shore wind farm project.

To answer the key question concerning local acceptance of the potential, near-shore wind farms, “what is your overall attitude towards the plans for near-shore wind farms in your local area?”, respondents were given answer-options on a 6 item Likert scale. Likert items ranged from “very negative” to “very positive”, and finally to “I don’t know”. This was combined with an open-ended question providing respondents with the opportunity to elaborate. A similar grid question asked respondents what Denmark as a society should do in order to reduce CO^2-emissions. Possible activities, ranging from investments in differing specific green-tech solutions to taxation regimes, were proposed. The section specifically on the WCS started with a quick explanation of the legal requirements of the scheme. A realistic wind farm share purchase price, and an estimated annual return of that share, was described, and the tax-free limit of 7000 kr. for WCS-investment return was mentioned. Respondents were then asked if they would consider investing in the WCS granted the opportunity to do so.

4. Results

In the following sections, we present some central data results via descriptive statistics. We look at 1) selected demographic factors and assess their influence on willingness to invest in the WCS, 2) consider the appeal of the WCS among wind farm opponents, and we present a regression model testing effects of independent variables on willingness to invest in the WCS. Finally 3), drawing on insights from the field of distributive justice (Walker, 2012), we discuss the extent to which the WCS is actually considered as compensation for grievances caused by potential near-shore wind farm projects by project stakeholders.

4.1. WCS: demographics matter

The opening question asked if respondents had previous knowledge of the WCS before answering the survey. 66.8% of the sample population (N = 1913) did not know of the WCS prior to receiving the survey invitation, but 33.2% were aware of the scheme. The empirical data reveals a significant willingness to invest in wind farm shares among respondents: 38.8% of the full sample would consider investing in the WCS, 31.9% of respondents would not, while 29.2% chose the option “I don’t know”.[6] Indicated attitudes towards the specific near-shore wind farm projects emerge as critical for willingness to invest in the WCS. As we will see, however, other factors influence the general appeal of the wind farm co-ownership scheme significantly too.

4.1.1. Gender

The effect of gender on willingness to invest is significant (see Fig. 2): The male sample population demonstrate almost 10% greater willingness to invest in the WCS than the female sample population. In economics it has been well documented that, all other things being equal, females appear somewhat more risk averse than men (Barber and Odean, 2001; Heckman et al., 2009). This tendency is also evident in the regression analysis presented later (see Table 1). Overall, a significant gender bias in general appeal of the WCS can be confirmed.

4.1.2. Age

15.5% of respondents picked the option “that I was younger” as a potential motivating factor for investing in the WCS, and among the respondents who would not invest in the WCS, 19.2% chose the option “I am too old to see the results of my investment” as the reason for that choice. During sales of shares follow-up explorative interviews on local perceptions of the WCS, and in the open-ended survey comments, older informants also underscored advanced age as a demotivating investment factor. In other words, age matters for willingness to invest in the WCS (Fig. 3).

4.1.3. Income

7.1% of respondents indicating that they would not invest in the WCS chose: “I cannot afford it right now” as the reason. If the money you have to spare every month when the bills are paid and groceries bought is very little, arguing that “I don’t invest in principle” (12.1%), that “I don’t understand the potential benefits” (12.6%), that “it is too complicated because of rules, regulations and taxes” (11.9%) or that “I have everything I need” (5.4%) may also be easy and logical choices. And they may very well be true. But as investment activities do require available funds for investment, or prioritization of the little extra money there is to spare into investment matters, for many low income families this will not be an option (Fig. 4).

4.2. To invest or not to invest: Windfarm perception matters

41.2% of the respondents considering investing in wind farm shares indicated that they would be doing so in order to support renewable/green energy, while 30.4% indicated their motivation for investment as potential economic gain. 13.2% chose support for local initiatives as their overall motivating factor. All respondents were asked what they anticipated might enhance their motivation for buying shares in the future. 8.9% of respondents anticipated more willingness to invest if a local company owned the turbines, 8.9% of the sample chose that ownership by a respectable company with a good reputation would enhance their willingness to invest, whereas 9.8% indicated that possible influence on the wind farm projects would increase their motivation for investing. So almost 30% of the respondents chose options reflecting ownership structures, influence and company reputation as potential motivators for investing in the WCS.

Fig. 5 shows that respondents already positive towards the potential, local wind farms are much more willing to invest in the WCS than respondents who are negative towards the potential wind farms. Curiously, however, a few “negative” or “very negative” respondents (n = 126) would consider investing in the WCS. We return to this phenomenon in the discussion. What is evident here is that the WCS does not immediately appeal to wind farm project opponents, however.

[6] This category is coded missing for subsequent analysis.

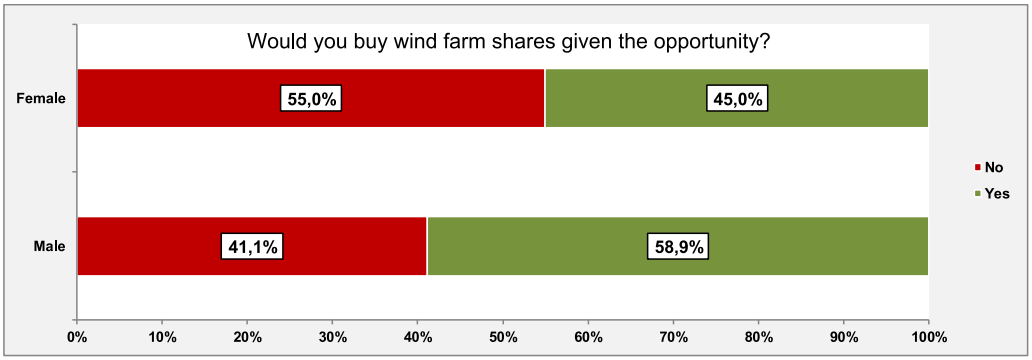


Fig. 2. Willingness to invest in wind farm shares by gender, $n = 1353$, $^{***}p < 0.001$.

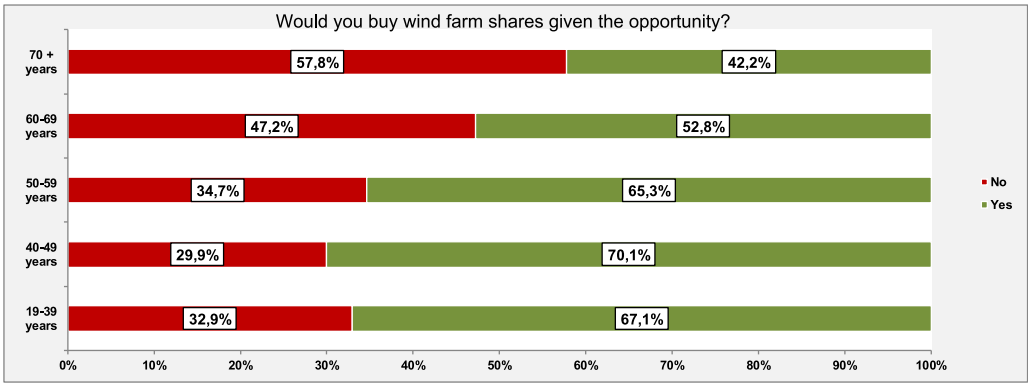


Fig. 3. Willingness to invest in wind farm shares by age in age categories, $n = 1220$, $^{***}p < 0.001$.

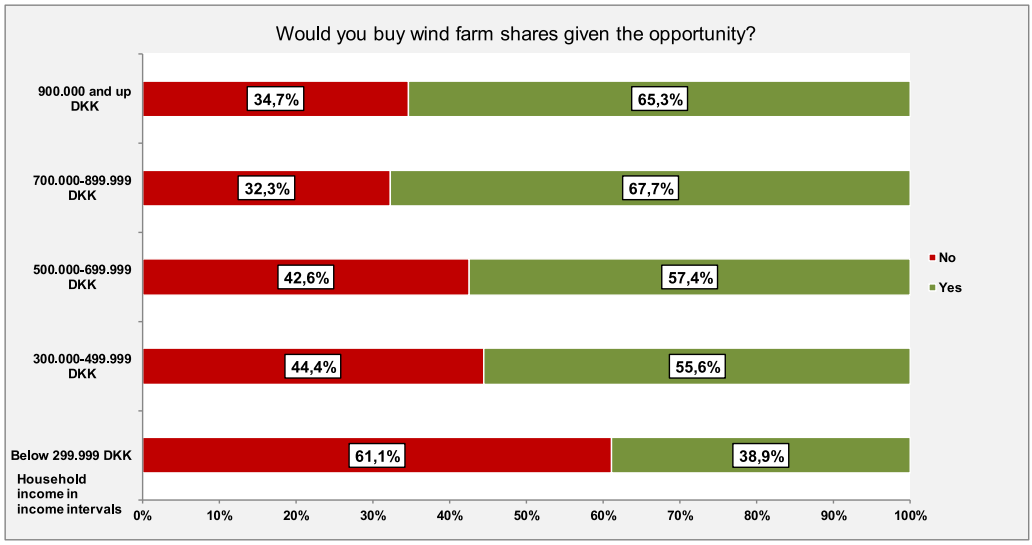


Fig. 4. Willingness to invest in wind farm shares by self-reported annual household income in Danish kroner, DKK, $n = 1202$, $^{***}p < 0.001$.

4.3. WCS: regression analysis

A three-step binomial logistic regression analysis assessed the effects of different variables on our dependent, the likelihood to buying shares via the WCS. All models shoved no evidence of multicollinearity as assessed by tolerance values greater than O.1.

Model 1 predictors were purely demographic variables: gender, age, education, income and residency type (permanent resident, PR, or

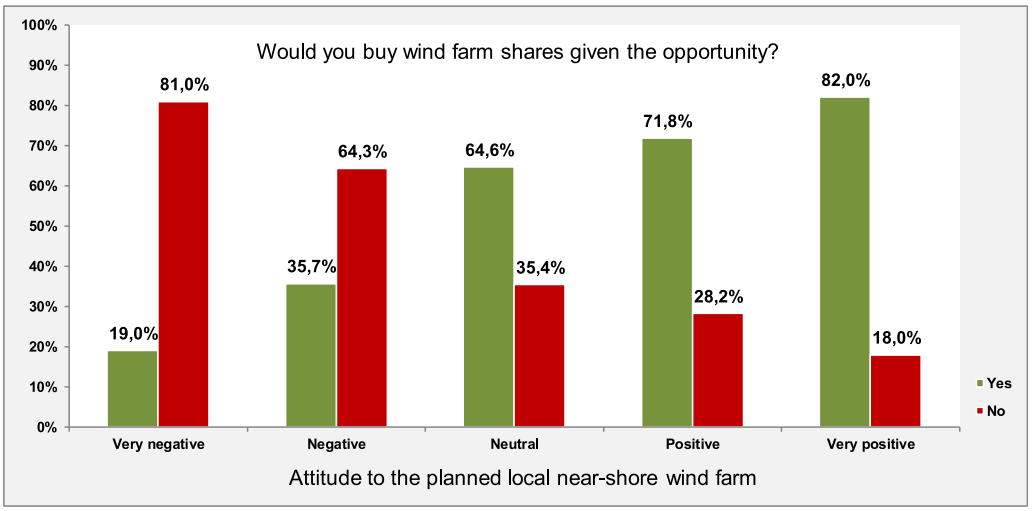


Fig. 5. Willingness to invest in wind farm shares by indicated attitude to the planned local near-shore wind farm, $n = 1349$, $^{***}p < 0.001$.

summerhouse owner, SHO). In model 2, we supplemented with the independent variable attitude towards wind power as a part of the Danish CO_2 reduction strategy, general support for wind power in Denmark. In model 3, we added indicated acceptance of the specific local near-shore wind farms. Judging by χ^2 tests all three models proved statistically significant.

A value of 0.092 Nagelkerke R^2 indicated 9.2% explained model variance. Overall, the model prediction success was 63.3%. Gender ($p < 0.001$), age ($p < 0.001$) and annual household income ($p = 0.001$) added significantly to the model/prediction, but levels of self-reported education ($p = 0.942$) did not. All other things being equal, in model 1 the effect of residency type is insignificant ($p = 0.163$) in relation to other demographic factors. Odds ratios < 1 tell us that females are less likely to invest in the WCS than males, and that willingness to invest declines with age (odds < 1). As levels of income increase so does willingness to buy shares (odds = 1.104). In model 2, we add indicated attitudes towards wind power in Denmark. This leads to explained Nagelkerke R^2 model variance of 25.1%. Statistical significance of demographic factors remains. In other words, demographics still matter for willingness to buy wind farm shares. Overall attitude to wind power is statistically significant ($p < 0.001$). Furthermore, odds ratios of 1.729 tell us that higher degrees of general support for wind power as CO^2 reduction strategy in DK positively, and quite significantly, influences the willingness to buy wind farm shares. Finally, moving on to model 3, we add indicated potential near-shore wind farm acceptance levels. Including specific project acceptance in the model Nagelkerke R^2 explained variance rises to 40%. In this model, statistical significance of local wind farm acceptance is strong ($p < 0.001$). The relative importance of general attitudes towards wind power decreases from odds ratios of 1.73–1.17, however. As acceptance levels of the potential near-shore wind farms increase, so does the indicated willingness to invest in shares via the WCS. With an odds ratio value of 2.273 the impact of local near-shore wind farm project acceptance levels is drastic. In model 3, education-levels emerge as slightly statistically significant ($p = 0.059$). Higher degrees of education correlate positively with willingness to invest in the WCS (odds = 1.12). Interestingly, in model 3 residency type emerges as a statistically significant factor too. Controlled for other variables, SHOs are more inclined to invest in the WCS than PRs are (odds = 1.58). Generally, what we see is strong evidence that especially support for the specific wind farm project plans significantly enhances the odds of investing in wind farm shares via the WCS. So while multiple real-world facts, such as economic liquidity, gender, age and general support for wind power as a national CO^2 reduction strategy all influence the willingness to invest in the WCS, perceptions of the specific local wind farm projects are crucial.

4.4. WCS: the moral dilemma

Answers/comments from open-ended answer options in the complete sample were coded in broad, descriptive categories through an iterative coding process. Of the 288 qualitative contributions on possible WCS investment motivation 80 respondents specifically mentioned geographical project proximity as problematic (see Lindén et al., 2015) for them. General comments on “[placing turbines] other places far out at sea”, or just blatantly “somewhere else”, are prevalent. 36 respondents specifically mentioned concerns about local nature/environment/populations, and in various colorful ways 48 respondents simply denied any possible present *or* future interest in the wind farm co-ownership scheme. Some respondents contributed with concrete suggestions for alternative wind farm locations near neighbouring towns. Of the 218 open-ended comments elaborating why respondents would not invest in WCS 130, this is more than half, explicitly touched upon the moral dilemma of investing in something you do not support in principle. The essence of the moral dilemma is captured in this comment: "I am against the construction of the wind farms, so it would be hypocritical and wrong [to invest in them]”. To some respondents the WCS was seen as bribery or as false compensation. Note this colorful remark: “The small consolation prices and ‘shut-up candy’[7] from the wind industry and the politicians do not interest me. I do not advocate bribery. There is no sensible reason to pollute the landscape with wind-turbine junk". Respondent interpretations of community benefit-schemes as bribery mirror some of the findings in related research (Aitken et al., 2016; Cass et al., 2010; Cowell et al., 2011). The following illustrative statement describes what is feared lost from the wind-turbines: “Economic benefit has nothing to do with the lost quality of life I believe I will experience if large wind-turbines harm nature, peace and tranquility at the coast". Note the direct emphasis on and explicit description of parallel value-sets at play here: Expected grievances by this respondent “[have nothing to do with] economic benefit”, he says.[8]

Turning to the metaphor of the webs of personalized significance

[7] The ironically labeled ‘hold kæft bolsje’ is a piece of candy so large that the child eating it cannot say anything, let alone protest. 'Hold kæft' means shut up in Danish.

[8] Interestingly, in the study by Cowell et al. on community benefits and windfarms, individual respondents asserted that: “no amount of money can compensate for the loss of the Welsh landscape” (2011, p. 551). The words used are almost identical to the above.

Table 1
Dependent variable: Indicated willingness to invest in the WCS. Coding: No, I would not invest in the WCS given the opportunity = 0. Yes, I would invest in the WCS given the opportunity = 1. M = Male, F = Female. PR = Permanent Resident, SHO = Second Home owner. Scales: Support for wind power as a carbon reduction strategy onshore and offshore combined: from very negative to very positive. Attitude towards the planned local near-shore wind farms: from very negative to very positive.

	Model 1			**Model 2**			**Model 3**		
	Logit coefficient	Standard error	Odds Ratio	Logit coefficient	Standard error	Odds Ratio	Logit coefficient	Standard error	Odds Ratio
Constant	1.44***	0.48	4.22	− 1.53***	0.59	0.22	− 2.44***	0.65	0.09
Demographics									
Gender. Coding: M = 0, F = 1	− 0.55***	0.15	0.58	− 0.64***	0.16	0.53	− 0.81***	0.17	0.45
Age	− 0.03***	0.01	0.97	− 0.02***	0.01	0.98	− 0.03***	0.01	0.97
Education	0.004	0.05	1.00	0.041	0.05	1.04	0.111*	0.06	1.12
Income	0.10***	0.03	1.10	0.11***	0.03	1.11	0.14***	0.04	1.15
Type of Residence. Coding: PR = 0, SHO = 1	− 0.20	0.15	0.82	− 0.09	0.16	0.91	0.46**	0.18	1.58
Attitude towards wind power generally									
Overall attitudes towards wind power				0.55***	0.05	1.73	0.16**	0.06	1.17
Local wind farm acceptance									
Local wind farm acceptance							0.82***	0.07	2.27
Model summary									
Model χ^2			73,759, $P < 0.01$			211,975, $P < 0.01$			362,009, $P < 0.01$
df			5			6			7
Pseudo R^2 (Nagelkirke)			0.09			0.25			0.40
Model prediction success			63.30			69.20			75.20
n =			1057			1045			1042

Notes:
* Significant at the 0.1 level.
** Significant at the 0.05 level.
*** Significant at the 0.01 level.

(Geertz, 1973; Nelson, 1996) described earlier, for this respondent "economic benefit" runs along one thread in the totality of his mental value-webs. Values inherent in "nature, peace and tranquility at the coast", on the other hand, run along a very different thread in those his complete webs of significance (Bidwell, 2013; Geertz, 1973). In this person's intricate personalized webs of human significance, then, the two divergent value-sets do not connect. And based on his statement it is unlikely they will ever meet – relative to this specific case and context, at least. In other words, the respondent does not expect that the WCS will ever compensate completely for expected grievances imposed by the potential nearshore wind farms. Possible WCS compensational-value and re-distributive potential simply speaks to the wrong place in this person's mental navigational map (Nelson, 1996) - even if some degree of economic distributive justice (Goedkoop and Devine-Wright, 2016; Walker, 2012) could potentially be achieved via the WCS in principle. So, the open-ended comments do provide evidence that some survey respondents do not feel justly compensated, or compensated in the right place, via the WCS. The practical complexity of community benefit and co-ownership schemes revealed in related research (Aitken et al., 2016; Cass et al., 2010; Cowell et al., 2011) is evident.

5. Discussion

5.1. WCS: pros, cons and what to be mindful of

The next sections summarize and discuss the presented quantitative and qualitative data. More specific comments and policy implications for the WCS and similar CBSs follow.

The presented empirical evidence suggests that 1) most potential WCS-investors already support the proposed wind farm projects, and that most project opponents will not engage in something they "are against in principle". Consequently, the overall mitigating effect of the WCS up front can be questioned. But data also suggests that a small group of project opponents may still be interested in WCS-investments (see Fig. 5). For this small group of otherwise very negative project stakeholders, then, the WCS may just have a positive effect in terms of greater local project engagement and acceptance long-term (see Musall and Kuik, 2011).

2) Demographic facts influence the general appeal of the scheme (see Figs. 2, 3 and 4). Willingness to engage in WCS investment activities decreases with age, and more importantly, the WCS is not equal for all: Those with fewer economic resources are actually subject to de-facto economic discrimination as the WCS presupposes investment liquidity. This is an unfortunate socioeconomic bias. So concerning the potential redistributive 'reach' of the WCS, we can conclude that socioeconomic and demographic facts have implications for the real-world potential for distributive justice via the WCS.

5.2. Distributive justice vs. moral dilemma

WCS is a financial incentive structure mandated by law; it is a CBS aimed at promoting local engagement in and acceptance of local wind farms (Anker and Jørgensen, 2015; Olsen and Anker, 2014; Sperling et al., 2010, p. 5445). But the WCS can also be framed as a compensation for local wind farm impact (Kerr et al., 2017; Zaunbrecher and Ziefle, 2015); the scheme can be interpreted as a CBS *compensating* local populations for bearing the potentially deep-felt burdens of proximity (Lindén et al., 2015) to large-scale renewable energy infrastructure. So, it could be argued that the WCS represents a political, practical, and very convenient vision that some economic re-distribution to local populations, in the form of investment openings, magically turns locally perceived project wrongs right.

Positive or negative framing regardless, the WCS *does* represent a very real and positive policy attempt at compensating local citizens for impacts of large-scale construction projects that were, in reality, often imposed on them (National Economic and Social Council & SLR Consulting, 2015). Despite the inevitable complications and challenges of the scheme. 3) To some degree the WCS *may* compensate for locally induced wind farm grievances via the re-distribution potential project gains, i.e. wind farm returns earned via WCS-investments, in wind farm host communities (Bidwell, 2013; Musall and Kuik, 2011). In the language of distributive justice, then, through WCS economic "goods" are potentially redistributed as compensation for local "harms" imposed by local wind farms (see Walker, 2012). The problem is the moral dilemma described in Section 4.4.: According to what appears to be a widespread moral logic among project stakeholders one cannot receive a potential "good" (economic benefits) from something perceived as a "bad" (a planned local wind farm).

5.3. WCS: values matter

Finally, and most importantly, potential economic gain does not appear to compensate the core of what is feared lost by many project stakeholders. We have seen that values prompting local negative reactions/resistance towards potential and realized wind farms are, for the most part, not about money (Bidwell, 2013; Cowell et al., 2011, p. 551). WCS is an investment scheme rooted in an economic rationality of "goods" via financial benefits. So while the WCS may successfully achieve some degree of local project-related economic re-distribution, and hence some degree of distributive justice (Walker, 2012), the scheme compensates through an individualized economic logic that may not soothe where local wind farm induced grievances hurt (see Bidwell, 2013). In fact, as the presented data suggests, for many project stakeholders it appears that the WCS completely misses the target. Paraphrasing the informant cited above: economic benefit does not match the quality of life feared lost due to the wind farms. In other words, the WCS does not reach the core of the project related grievances for many project stakeholders, and personal values activated by economic benefit are (or can be) at a very different place than values activated, and potentially violated, by local wind farm projects. In this article, these seemingly incompatible personal value-sets have been illustrated via the metaphor of personalized webs of significance. Judging by data presented in this case-study, it seems that the immediate effect of the WCS as a CBS automatically prompting local social acceptance of planned wind farm projects should be questioned.

6. Conclusion and policy recommendations

But the WCS is only 1 out 4 Danish REA policy measures aimed at promoting social acceptance of wind farms in Denmark (Anker and Jørgensen, 2015; Energi-, Forsynings- og Klimaministeriet, 2017; Olsen and Anker, 2014). For the near-shore wind farms, arguably the WCS is the most significant of these policy measures, however. 1) The loss of property-value scheme is a true *compensation* scheme (Olsen and Anker, 2014), 2) the community benefit scheme/the green scheme did not apply to near- and off-shore wind farms, and 3) the guarantee fund supports local project initiatives via loan guarantees for preliminary project investigations (Anker and Jørgensen, 2015; Energinet.dk, n.d.-a). Importantly, the WCS is a CBS model that is potentially adaptable to multiple renewable energy contexts.

While policymakers should generally be mindful of the key demographic and empirical trends presented in this article, some supplementary, more practical remarks concerning the wind farm co-operative initiative implementation process, and the potential mediating effects of the scheme, are appropriate too.

The timing (Ellis and Ferraro, 2017; Goedkoop and Devine-Wright,

2016) of the WCS may influence the general appeal, and potential local reach, of the scheme. Currently Danish legislation demands that the wind farm shares are offered for sale locally before grid connection (Energinet.dk, n.d.b). Perhaps simply postponing or prolonging the time during which the wind farm shares are sold could increase local interest in the WCS for projects with fewer WCS investors. Alternatively, initiating a wind farm cooperative in the very early project planning phases could encourage even greater levels of local/community engagement and project acceptance via the co-operative.

The level of compulsion of the scheme, i.e., that it is an interventionist, statutory requirement mandated by law (Anker and Jørgensen, 2015; Kerr et al., 2017), may influence the level of 'local engagement' WCS generates. On the one hand, views have been observed that CBSs should be of a more "local and voluntary" character (Cowell et al., 2011, p. 553). On the other hand, especially among project opponents, voluntary, negotiated CBSs may well be interpreted all the more as local "bribes" (Cass et al., 2010). Accordingly, in related research the view that "a model (…) of significant formalized benefits in-kind appears more straightforwardly acceptable" (Cass et al., 2010, p. 272) has been observed among members of the public.

The inherent complexity of 'community' is of crucial importance for the possible effects of CBSs (Aitken et al., 2016; Cowell et al., 2011, p. 544), and our final argument touches upon precisely the notion of *community*. One could question the categorization of the WCS as a CBS: WCS benefits the individual investor with the funds to invest, and the potential wind farm investment returns are then re-distributed to that individual investor. Via the WCS, then, potential wind farm "goods" do not return to the *community*, despite the fact that the entire community, the municipality, bears the potential wind farm impacts (and burdens). Thus, it could be argued that the WCS cannot really be seen as a community benefit scheme at all. Perhaps some reasonable wind energy production dependent municipal level benefit scheme would facilitate a broader, more socially neutral community benefit for the entire community/wind farm host municipality.

6.1. Conclusion

In this article we have explored local perceptions of the Danish wind farm co-ownership initiative, WCS. Through the analytical lens of distributive justice (Walker, 2012), we have analyzed locally perceived redistributional effects of the co-ownership scheme. Empirical evidence suggests that 1) demographic facts, such as gender, age and income influence the general appeal of the scheme. Furthermore, the WCS is not equal for all: Engagement in wind farms via the WCS presupposes investment liquidity. 2) Most potential WCS-investors already support the proposed wind farm projects, while many project opponents would not engage themselves in something they "are against in principle" (Cowell et al., 2011). Consequently, the mitigating effect of the WCS can be questioned. Finally 3), potential monetary gain does not appear to justly compensate the core of what is actually feared lost by many wind farm project stakeholders – such as values related to "nature, peace and tranquility at the coast" (see Bidwell, 2013; Cowell et al., 2011). But that said, wind farm co-ownership via the WCS *does* promote some level of local engagement in wind farm projects among WCS investors (Musall and Kuik, 2011), and offering local wind farm co-ownership demonstrates willingness to share the potential wind farm "goods". So overall, the WCS is a very positive policy attempt at creating local engagement in local wind farms via wind farm co-ownership. And crucially, given "tensions [remaining] between vocal commitments to democratic principles within planning processes and concerns about the practical value and limitations of public participation" (Aitken et al., 2016, p. 559), the WCS is practically possible. As we have seen throughout this article, however, the WCS alone is not, and never will be, a tick the box exercise instantly resolving real and perceived burdens of wind farm proximity among wind farm host communities. Real world facts and complications, such as demographics, economic liquidity, preconceived project perceptions and personal values, get in the way.

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Appendix A

Appendix 1

Appendix 1

Selected sample demographics (I, II, III): from the survey sample that is drawn from within relative proximity to the coast (see notes below for details). Reference demographics (IV): municipal level data on adult residents that live permanently in the near-shore wind farm host municipalities (source: Statistics Denmark).

Household income by income intervals		**Below 299.999 DKK**	**300.000-499.999 DKK**	**500.000-699.999 DKK**	**700.000-899.999 DKK**	**900.000 DKK and up**
From the survey sample - drawn from close to the coast						
Survey sample - Full Sample	I	18%	27%	22%	18%	15%
Survey sample - Permanent Residents	II	23%	29%	24%	16%	8%
Survey sample - Second Home Owners	III	11%	25%	17%	21%	26%
Municipal level reference demographics						
Average at municipal level in the 5 near-shore wind farm host municipalities (source: Statistics Denmark)	IV	40%	26%	15%	11%	8%

Highest attained education (15-69 years)		**Up to high school or similar**	**Vocational training**	**Shorter education**	**Medium length education**	**Longer higher education**
From the survey sample - drawn from close to the coast						
Survey sample - Full Sample	I	11%	33%	9%	30%	17%
Survey sample - Permanent Residents	II	12%	37%	10%	28%	13%
Survey sample - Second Home Owners	III	7%	26%	9%	35%	24%
Municipal level reference demographics						
Average at municipal level in the 5 near-shore wind farm host municipalities (source: Statistics Denmark)	IV	40%	41%	3%	12%	3%

(*continued on next page*)

Appendix 1 (*continued*)

Household income by income intervals		Below 299.999 DKK	300.000-499.999 DKK	500.000-699.999 DKK	700.000-899.999 DKK	900.000 DKK and up
Age by age intervals in years		**19-39**	**40-49**	**50-59**	**60-69**	**70 +**
From the survey sample - drawn from close to the coast						
Survey sample - Full Sample	I	6%	16%	26%	33%	19%
Survey sample - Permanent Residents	II	10%	18%	25%	29%	18%
Survey sample - Second Home Owners	III	2%	13%	28%	38%	19%
Municipal level reference demographics						
Average at municipal level in the 5 near-shore wind farm host municipalities (source: Statistics Denmark)	IV	21%	16%	18%	18%	19%

Notes:

Sample demographics (I, II, and III): The *survey sample* was drawn from within relative proximity to the coast close to the DEA selected near-shore wind farm sites. The sample was drawn in this way to ensure a broad sample of respondents who lived in/owned properties fairly close to the near-shore wind farm sites of interest. The survey sample was *not* intended to be representative of municipal level demographics (or of the Danish population), but intentionally prioritized residents and RET stakeholders facing the planned RET projects much more locally.

In the final survey sample, approximately 40% of the respondents lived in/owned properties up to 500 meters from the coast, about 40% lived in/owned properties between 500 – 2000 meters from the coast, and less than 20% of the respondents lived in/owned properties more than 2000 meters from the coast.

Reference demographics (IV): The *reference demographics* presented are drawn from Statistics Denmark, and this data represents all adults that live permanently (the permanent area residents, PRs) in the near-shore wind farm *host municipalities*. This data, then, shows demographics at the municipal level, and therefore it also includes residents living further inland – far from the DEA selected near-shore wind farm sites. In these municipalities the areas close to the sea and to the coast do tend to be the most attractive and popular. These statistics from Statistics Denmark, and their overall implications vis-à-vis demographics in the near-shore wind farm host municipalities, may be of general interest to the reader.

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